

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data integration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks:20

Annexure A

CLASS TEST

Criteria	Understanding of Task (2)	Procedure & Execution (2.5)	Observation & Result (2)	Viva Voce / Conceptual Response (2)	Time Management & Presentation (1.5)	Total(10)
Weightage	20%	25%	20%	20%	15%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						
<i>Q6</i>						
<i>Q7</i>						
<i>Q8</i>						
<i>Q9</i>						
<i>Q10</i>						

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

CLASS TEST

QUESTIONS:

Total Marks: 10x2 =20

Q1. Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it. [L2, L4]

Q2. What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1, L2, L3]

Q3. Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration. [L2, L4]

Q4. What is covariance analysis? List the two covariance formulas [L2, L3]

Q5. Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Question: If the stocks are affected by the same industry trends, will their prices rise or fall together and independent ?

Q6. Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. [L2, L4]

Q7. Find covariance for following data set $x = \{4, 5, 6, 7, 9\}$, $y = \{8, 3, 7, 5, 6\}$

Q8. Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps. [L3, L4, L5]

Q9. A hospital is analyzing patient **age** data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency** (equi-depth) binning. **Sample Data Set (Ages):** 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]

Q10. Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples, and evaluate their impact on data analysis. [L3, L4, L5]

Co-1 Assessment tool-3

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1. Define Data Mining and list any two Basic data Mining tasks. Explain the steps involved in KDD Process and justify the role of data PreProcessing.
 2. What data cleaning? Mention two Common Data Quality issues. Explain methods to handle missing values and noisy data. example.
 3. Define Data integration and list two challenges associated. with it explain how Redundancy and inconsistency during data integration.
 4. What is the covariance Analysis? List the two Covariance formulae.
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- 5) Suppose two Stocks A and B have the following values in one week. If the stocks are affected by the same industry trends, together and ?
- 6) Explain the Architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze it.
- 7) Find Covariance for the following Data Set $X = \{4, 5, 6, 7, 9\}$, $Y = \{8, 3, 7, 5, 6\}$
- 8) Describe the Major steps in data Re-Processing. Give a Data set with missing values, outliers, and justify the Sequence of steps.
- 9) A hospital is Analyzing Patient age data to identify Board healthy trends. Because minor variations and noise Sample Data Set (Ages).
13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 45, 46, 52, 70, .
- 10) Explain data discretization and its importance in data mining. Compare equal-width and the equal-frequency binning with suitable examples and evaluate their impact on Data Analysis.

Following:- Data Mining:- The Process of discovering useful Patterns, relationship and knowledge from large datasets.

→ Two basic tasks:-

- a. classification.
- b. clustering

2
Sol:- Data cleaning:- The Process of detecting and Correcting inaccurate, incomplete (or) inconsistent Data.

Two data Quality issues:-

- a. Missing values
- b. Noisy (incorrect) data.

3
Sol:- Data integration:- Combining data from Multiple Sources in to single, consistent datasets.

Two challenges:-

- a. Data redundancy
- b. Data inconsistency

4
Sol:- → Covariance Analysis:- Measures how two variables change together.

→ Two Covariance formulas
a. Population Covariance

$$\sigma_{xy} = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$
$$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

5
Sol:- Yes. The stock Prices will rise or fall together because they have Positive Covariance, indicating they move

5) Components:-

1. Data Sources
2. Database/Data Warehouse
3. Data Mining Engine

4. Pattern Evaluation

5. User Interface.

These components work together to discover useful knowledge from data.

7
Q1:- $X = \{4, 5, 6, 7, 9\}$, $Y = \{8, 3, 7, 5, 6\}$

→ Mean of $X = 6.2$, → Sample Covariance

→ Mean of $Y = 5.8$,

$$S_{xy} = \frac{2.2}{4} = 0.55$$

Covariance = 0.55 (Positive)

8
Q1:- Major Steps:-

1. Data cleaning
2. Data integration
3. Data Transformation
4. Data Reduction
5. Data Discretization

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Q1:- Equal-frequency binning divides the sorted data into bins containing an equal number of values. Each value in a bin is replaced by the bin mean (or bin median) which reduces noise while preserving trends.

10
Q1:- → Data Discretization:- Converting continuous values into discrete intervals (bins)

Importance:-

- a. Reduces data complexity
- b. Improves mining efficiency

→ Methods:-

- Equal-width binning
- Equal-frequency (equi-depth) binning